

Research Statement

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I am an econometrician broadly interested in developing statistical and causal inference methods that leverage unstructured data sources—e.g., text, image, audio, video, location trace, or multi-modal data. My work draws from and extends tools in semiparametric and high-dimensional statistics as well as methods in AI and machine learning to deliver principled, general-purpose frameworks for econometric analysis with unstructured data. The applications of such work span many fields, but are especially relevant in political economics, development economics, and finance—fields in which unstructured (text and image) data are often critical for forming novel economic insights.

My graduate studies have focused on several distinct projects in support of this broad research focus, including: (1) my job market paper [Carlson, 2025], which develops a novel framework for making statistically and scientifically principled “discoveries” from unstructured datasets in an unsupervised way, accomplished by extending two frontier econometric frameworks and one popular AI interpretability framework; (2) a project that builds on insights from semiparametric inference for missing data to inform a general framework for robust and efficient inference with AI predictions based on unstructured data [Carlson and Dell, 2025]; (3) a project that develops a fully nonparametrically founded framework for dynamic causal inference with a single time series (the “potential system”), which directly motivates a role for flexibly modeled functions of unstructured data in time series causal inference [Carlson and Shephard, 2026]; and (4) a variety of projects that directly focus on the large-scale curation of new unstructured data sources, made possible with new innovations in AI/ML methods [Dell et al., 2023, Carlson et al., 2024a, Bryan et al., 2023, Shen et al., 2021]. I have also worked on econometrics projects that treat topics in misspecification and clustered standard errors, respectively [Andrews et al., 2026, Carlson et al., 2024b].

My job market paper, “Making Interpretable Discoveries from Unstructured Data: A High-Dimensional Multiple Hypothesis Testing Approach” [Carlson, 2025], proposes a framework for making rich and interpretable discoveries (rejections of interpretable null hypotheses, formally) from unstructured datasets, offering a more statistically and scientifically rigorous alternative to conventional qualitative coding practices [Haaland et al., 2024]. At its core, the framework maps unstructured data points to high-dimensional, interpretable “concept embeddings” and then analyzes the resulting dataset using new results and tools in high-dimensional multiple hypothesis testing. The framework is made possible by extending two frontier econometric frameworks: the high-dimensional “many approximate means” framework of Belloni et al. [2018] and the generalized FWER controlling framework of Romano and Wolf [2007]. The framework also provides a novel statistical formalization of the “detection scoring” AI interpretability framework of Paulo et al. [2024], showcasing how econometric insights may be readily applied to AI evaluation workflows. The framework is applied to a variety of both descriptive and causal

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analyses in prominent empirical economics papers, recovering original discoveries and upturning new interpretable discoveries as well.

Other times a researcher is not interested in unsupervised discovery from unstructured data, but instead in leveraging predictions of known quantities of interest (e.g., based on theory or previous empirical studies) in downstream econometric analyses. In joint work with Melissa Dell, “A Unifying Framework for Robust and Efficient Inference with Unstructured Data” [Carlson and Dell, 2025], we propose a simple perspective and econometric framework for robust and efficient inference in such a setting: that predictions from unstructured data using deep neural networks can be viewed as imputations of low-dimensional missing structured data, and, as such, statistical methods for missing data (including many of those for causal inference) can be applied under appropriate adaptation. With this insight, we develop doubly robust and semiparametrically efficient estimators of parameters defined by moment restrictions based on predictions from unstructured data, nesting the results of other popular frameworks along the way [Angelopoulos et al., 2023], and further generalize to settings where predictions from unstructured data are aggregated before being analyzed in downstream linear regressions.

The semi/nonparametric identification results for causal parameters that motivate AI/ML predictors in cross-sectional studies are largely absent in studies of pure (single) time series. This is because most modern efforts in causal inference for a single time series (such as in macroeconometrics) are parametrically founded or projection focused; other nonparametric causal inference efforts exist, but largely focus on panel data, and estimands of interest typically differ from the single time series setting. To bridge this gap in the literature, in joint work with Neil Shephard, “When are time series predictions causal? The potential system and dynamic causal effects” [Carlson and Shephard, 2026], we develop a nonparametric, potential outcomes-style foundational model for dynamic causal inference with a single time series: a time series system we call the “potential system.” Using the potential system, we derive novel but familiar looking estimands and nonparametric identification results for dynamic causal parameters in a single time series, and thereby motivate time series applications for causal machine learning tools and techniques long used in the cross-sectional causal inference literature. Such results, in turn, directly suggest how unstructured data may be leveraged in identification and estimation of dynamic causal effects.

Pure AI/ML methods development for curating new sources of unstructured data was the subject of several projects at the beginning of my graduate studies. In “Efficient OCR for Building a Diverse Digital History” [Carlson et al., 2024a], joint work with Melissa Dell and Tom Bryan (and an ACL *oral presentation*), we develop a new neural architecture for performing optical character recognition (OCR) in low-resource languages, based on the insight that, in the interest of sample efficiency, OCR can be treated as an information retrieval problem in a particular metric space. This OCR method was later adopted in “American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers” [Dell et al., 2023], helping enable the digitization of over 20 million scans in the Library of Congress’s public domain Chronicling America collection.

I plan to continue centering econometric methods for unstructured data in my research agenda, soon expanding into work that considers unstructured data as treatment: both simplifying causal inference based on latent treatments exposed from unstructured data, and optimizing generative AI to deliver unstructured (text or image) treatments with quantifiably large causal effects. I am further interested in extending many of the frameworks I have developed already to address both longstanding questions related to monetary policy causal effects and nascent questions in economic measurement via classification of AI usage data.

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